Chain Rule for partial derivatives.

Let $z = f(x, y)$ $\quad x = x(u, v)$ & $y = y(u, v)$

$\Rightarrow z = f\big(x(u,v),\, y(u,v)\big)$

$y = f(x)$

$\frac{dy}{dx} = f'(x)$

$\frac{\partial z}{\partial u}$ $\quad\quad \frac{\partial z}{\partial v}$

If $x = x(u, v)$ & $y = y(u, v)$ have first order derivatives at (u, v) and if $z = f(x, y)$ is differentiable at $(x, y) = \big(x(u,v),\, y(u,v)\big)$, then $z = f\big(x(u,v), y(u,v)\big)$ has first order partial derivatives at (u, v) given by

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial z}{\partial v}$$

$$\begin{bmatrix} \partial z/\partial u \\ \partial z/\partial v \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{bmatrix} \begin{bmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{bmatrix}$$

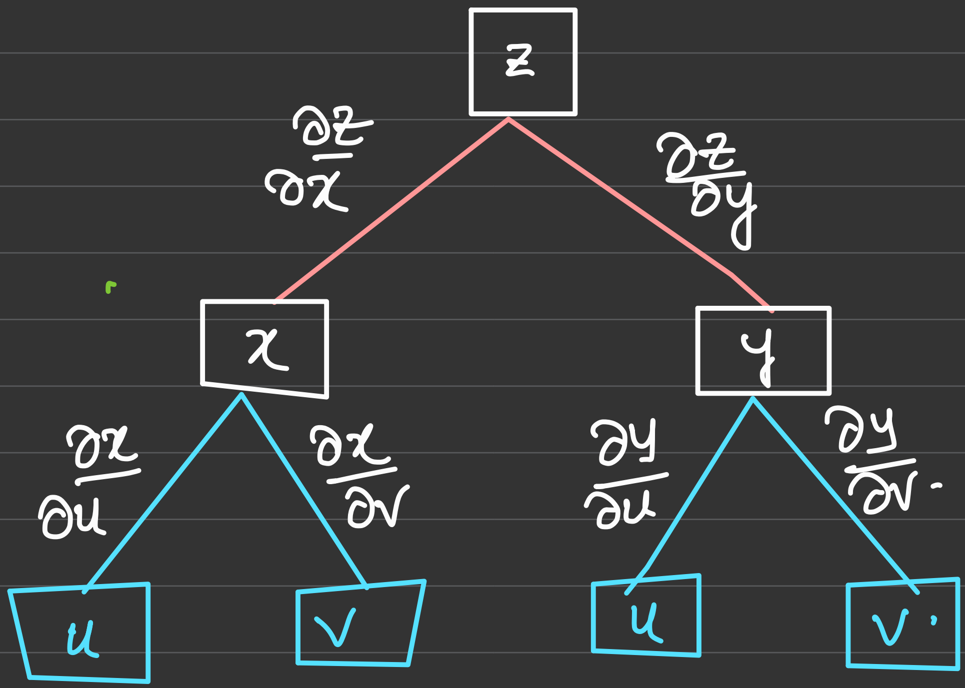
* $x = x(u,v)$, $y = y(u,v)$ $z = z(u,v)$. → first order partial derivatives exist at (u,v).

$$w = f(x,y,z).$$
$$= f\big(x(u,v),\ y(u,v),\ z(u,v)\big)$$

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v}$$

$$z = f(x,y) = f\big(x(u,v),\ y(u,v)\big)$$



$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}.$$

Ex: $z = e^{xy}$ $\quad x = 2u+v \quad y = \frac{u}{v}$.

find $\frac{\partial z}{\partial u}$ $\quad \frac{\partial z}{\partial v}$

Soln: $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$

$= ye^{xy}(2) + xe^{xy}\left(\frac{1}{v}\right)$. $\quad = \left(2y + \frac{x}{v}\right)e^{xy}$

$= \left(\frac{2u}{v} + \frac{2u+v}{v}\right)e^{(2u+v)\frac{u}{v}}$

$= \left(\frac{4u}{v} + 1\right)e^{(2u+v)(u/v)}$.

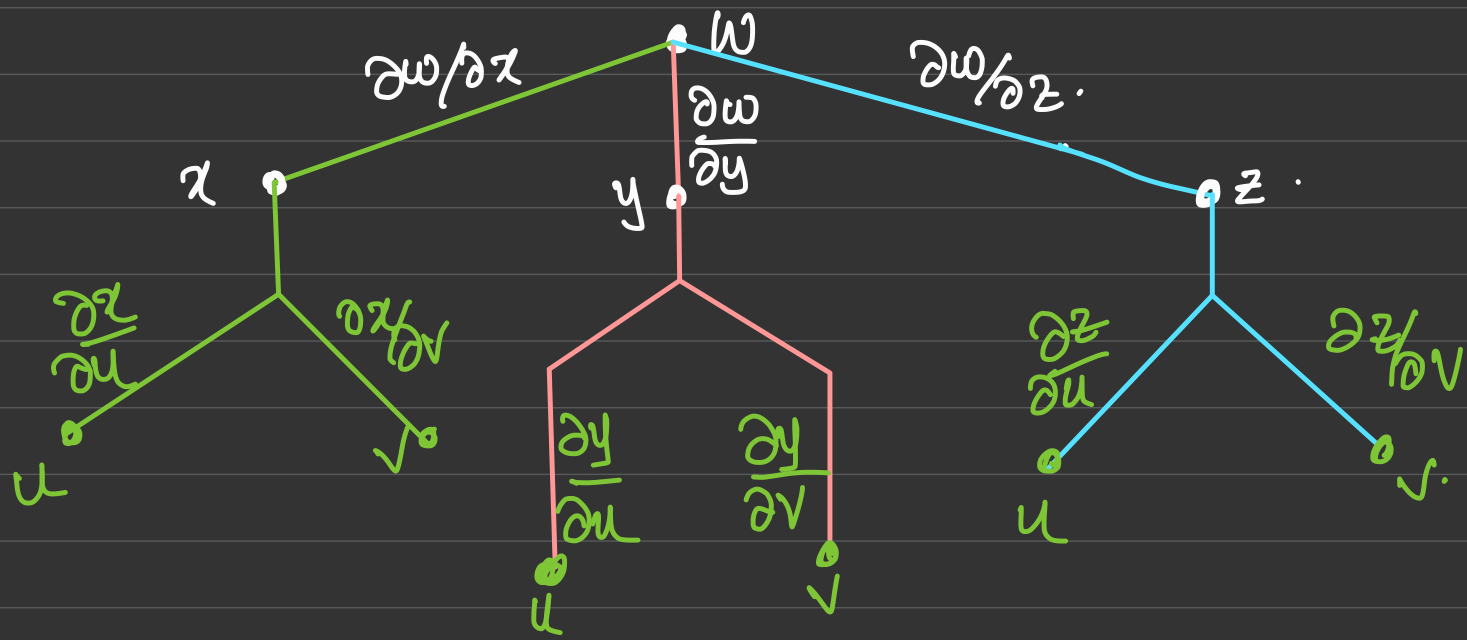
$\frac{\partial z}{\partial v} = -2\left(\frac{u}{v}\right)^2 e^{(2u+v)(u/v)}$.

$w = e^{xyz}$ $\quad x = 3u+v$

$y = 3u-v$

$z = u^2v$.

$\frac{\partial w}{\partial u}$ $\quad \frac{\partial w}{\partial v}$



In general if $w = f(v_1, v_2 \ldots v_n)$ of n variables, and suppose each v_i is a fn of t, then

$$\frac{dw}{dt} = \frac{\partial w}{\partial v_1}\frac{dv_1}{dt} + \frac{\partial w}{\partial v_2}\frac{dv_2}{dt} + \cdots + \frac{\partial w}{\partial v_n}\frac{dv_n}{dt}.$$

$$\rho^2\sin^2\phi(\cos^2\theta+\sin^2\theta)-\rho^2\cos^2\phi$$

Ex: $w = x^2+y^2-z^2 \to \rho^2\sin^2\phi\cos^2\theta+\rho^2\sin^2\phi\sin^2\theta-\rho^2\cos^2\phi$

$x = \rho\sin\phi\cos\theta$, $\quad y = \rho\sin\phi\sin\theta \quad z = \rho\cos\phi$

find $\frac{\partial w}{\partial\rho}$, $\frac{\partial w}{\partial\theta}$.

$$\rho^2\sin^2\phi-\rho^2\cos^2\phi$$

$$w = -\rho^2\cos 2\phi.$$

$$\frac{\partial w}{\partial\rho} = -2\rho\cos 2\phi \qquad \frac{\partial w}{\partial\theta} = 0 \qquad \frac{\partial w}{\partial\theta} = 0.$$

Ex. Let $t = \frac{u}{v}$ $\quad u = x^2-y^2 \quad v = 4xy^3$ find $\frac{\partial t}{\partial x}$ & $\frac{\partial t}{\partial y}$.

$$\frac{\partial t}{\partial x} = \frac{\partial t}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial t}{\partial v}\frac{\partial v}{\partial x} = \frac{1}{4xy^3}2x+\left(\frac{-u}{v^2}\right)(4y^3)$$

$$= \frac{2x}{4xy^3}+\frac{(y^2-x^2)}{(4xy^3)^2}(4y^3)$$

$$= \frac{2x(4xy^3)+(y^2-x^2)(4y^3)}{(4xy^3)^2} = \frac{\cancel{4y^3}(x^2+y^2)}{4xy^{\cancel{3}}.4xy^3}$$

$$= \frac{x^2+y^2}{4x^2y^3}.$$

$$\frac{\partial t}{\partial y} = \frac{(y^2-3x^2)}{4xy^4}.$$

* Find the velocity at $t = \frac{\pi}{4}$ given the position fn. Also find the speed after $t = \pi/4$.

$$\vec{r}(t) = (3, 2t, \sin t) = (f_1(t), f_2(t), f_3(t))$$

$f_1(t) = 3$; $f_2(t) = 2t$, $f_3(t) = \sin t$.

Velocity $\vec{v}(t) = \left[\frac{df_1(t)}{dt} \quad \frac{df_2(t)}{dt} \quad \frac{df_3(t)}{dt}\right]$

$$= [0, 2, \cos t] \quad \text{at } \pi/4$$

$$= \left[0, 2, \frac{1}{\sqrt{2}}\right]$$

Speed $= \|\vec{v}(t)\| = \sqrt{2^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$

$= \sqrt{4 + 1/2} = \sqrt{9/2}.$

Ex: $w = 5\cos xy - \sin xz$ $\quad x = \frac{1}{t},\ y = t,\ z = t^3.$

Find $\frac{dw}{dt}$. =

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$$

UNIT: 3 $\rightarrow$ Kalman-filter

$\rightarrow$ GMM

EM algorithm

WLS $\rightarrow$ WLS.

RLS.

08.02.2026.

Directional Derivatives & Gradients.

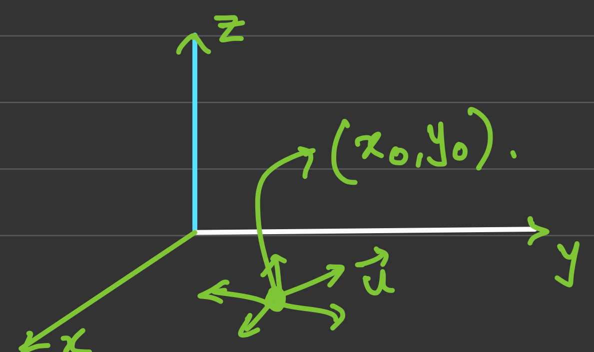
Let $z = f(x,y)$

$$\frac{\partial z}{\partial x} = f_x(x,y) \qquad \frac{\partial z}{\partial y} = f_y(x,y).$$

$$\left.\begin{aligned} f_x(x,y) &= \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}. \\ f_y(x,y) &= \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}. \end{aligned}\right\}$$

Rate of change of $f(x,y)$ in direction parallel to x & y-axis.

We want to look at the rate of change of $f(x,y)$ in other directions.



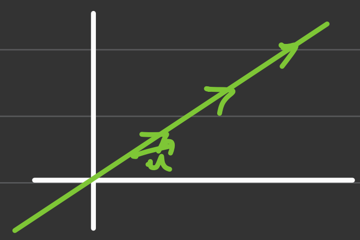
Partial derivatives of $f(x,y)$ gives the instantaneous rates of change of $f(x,y)$ in directions parallel to the coordinate axes

If we want to compute the rate of change of $f(x,y)$ with respect to distance in any direction, we use directional derivatives.

$f(x,y) \rightarrow$ Compute the instantaneous rate of change of $f(x,y)$ w.r.to distance from a point (x_0, y_0) in some direction.

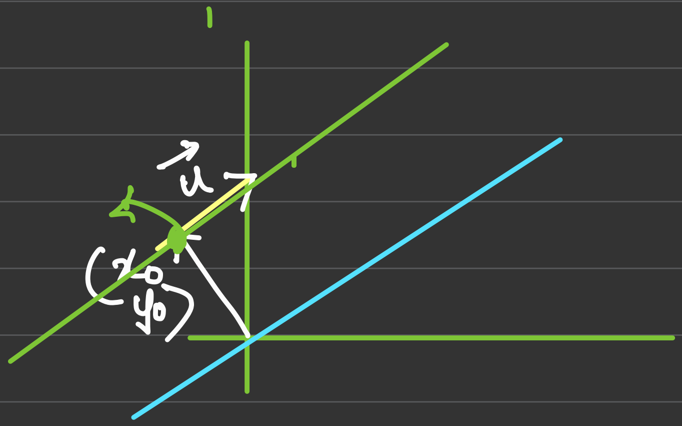
There are infinitely many directions along which we could move from (x_0, y_0). Let $\vec{u}$ be a unit vector along whose direction we want to move.

$$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \leftarrow$$



$$\overbrace{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + s\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}}.$$

$$= \quad \vec{v} + s\vec{u}$$



$$z = f(x_0 + su_1,\ y_0 + su_2).$$

$$= f\left(\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + s\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right)$$

$\dfrac{dz}{ds}$ at $s = 0$ $\rightarrow$ Instantaneous rate of $f(x,y)$ w.r. to the distance from $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ along $\vec{u}$.

Directional Derivative $D_{\vec{u}} f(x_0, y_0)$

If $f(x,y)$ is a function of (x,y) & $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ is a unit vector, then the directional derivative of f in the direction of $\vec{u}$ at $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ is given by

$$D_{\vec{u}} f(x_0, y_0) = \frac{d}{ds}\left[f(x_0 + su_1, \; y_0 + su_2)\right]\Big|_{s=0}.$$

provided the derivative exists.

$D_{\vec{u}} f(x_0, y_0)$ is the slope of the surface $z = f(x,y)$ in the direction of $\vec{u}$ at the point $(x_0, y_0, f(x_0, y_0))$

$D_{\vec{u}} f(x_0, y_0)$ depends on (x_0, y_0) & the direction $\vec{u}$.

Ex: $f(x,y) = xy$, $(x_0, y_0) = \underline{\underline{(1,2)}}$. Obtain the $D_u f(1,2)$ in the direction of $\vec{u} = \begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}$

$$D_{\vec{u}} f(x_0, y_0) = \frac{d}{ds}\Big[f(x_0 + su_1,\ y_0 + su_2)\Big]_{s=0}.$$

$$= \frac{d}{ds}\Big[f\Big(1 + s\sqrt{3}/2,\ 2 + s/2\Big)\Big]_{s=0}.$$

$$f\left(1 + \frac{s\sqrt{3}}{2},\ 2 + \frac{s}{2}\right) = \left(1 + \frac{s\sqrt{3}}{2}\right)\left(2 + \frac{s}{2}\right)$$

$$= 2 + \frac{s}{2} + s\sqrt{3} + \frac{s^2}{4}\sqrt{3}$$

$$f(x_0 + su_1,\ y_0 + su_2). = \frac{\sqrt{3}s^2}{4} + s\left(\sqrt{3} + 1/2\right) + 2.$$

$$\frac{d}{ds}\left(f(x_0 + su_1,\ y_0 + su_2)\right) = \frac{\sqrt{3}s}{2} + \left(\sqrt{3} + \frac{1}{2}\right)\Big|_{s=0} \Rightarrow \sqrt{3} + \frac{1}{2}. \approx 2.232$$

Conclusion: If we move a small distance from $(1,2)$ in the direction of $\vec{u} = \begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}$, $f(x\,y) = xy$ will increase by about 2.23 times the distance moved.

* Suppose f is a function from $\mathbb{R}^3 \longmapsto \mathbb{R}$, let $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$ be a unit vector, and f is a fn $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$, then

$$D_{\vec{u}} f(x_0, y_0, z_0) = \frac{d}{ds}\Big[f(x_0 + su_1,\ y_0 + su_2,\ z_0 + su_3)\Big]_{s=0}.$$

provided the derivative exists.

$f: \mathbb{R}^2 \to \mathbb{R}$. $\quad z = f(x,y)$. $\quad D_{\vec{u}} f(x_0, y_0)$, $\quad \vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$

$z = f(x, y)$

$$\underline{z = f(x_0 + s u_1,\ y_0 + s u_2)} \Rightarrow$$

$x = x(s) = x_0 + s u_1$, $\quad y = y(s) = y_0 + s u_2$

Applying chain rule

$$D_{\vec{u}} f(x_0, y_0) = \frac{d}{ds}\Big[\overbrace{f(x_0 + s u_1,\ y_0 + s u_2)}^{f(x,y)}\Big]_{s=0} \qquad x = x_0 + s u_1,\ y = y_0 + s u_2$$

$$= \left.\frac{dz}{ds}\right|_{s=0} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$= f_x(x_0, y_0)\, u_1 + f_y(x_0, y_0)\, u_2$$

$$= \begin{bmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{bmatrix} \begin{bmatrix} \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial s} \end{bmatrix}$$

$w^T v \Rightarrow \vec{w} \cdot \vec{v}$

$$= \begin{pmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{pmatrix}^T \begin{pmatrix} \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial s} \end{pmatrix}$$

$$= (\nabla z) \circ \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$= \nabla z \cdot \vec{u}$$

$$D_{\vec{u}} f(x_0, y_0) = \underline{\underline{\nabla f(x_0, y_0)}} \circ \vec{u}.$$

$\vec{u}$: Unit vector along which we seek the directional derivative.

$\Downarrow$

Slope of $f(x,y)$ at (x_0, y_0) along $\vec{u}$

$$D_{\vec{u}} f(x_0, y_0) = \|\underline{\underline{\nabla f}}\| \, \|\vec{u}\| \cos\theta.$$

$\theta = 0°$

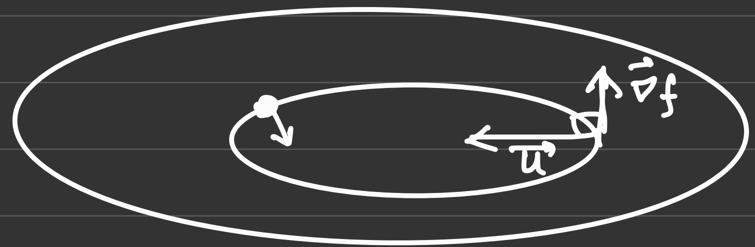
$\theta = 90°$

$\underline{\underline{D_{\vec{u}} f(x_0, y_0)}} = 0$ if $\theta = \underline{\underline{90°}}$ $\rightarrow$ Level curve.

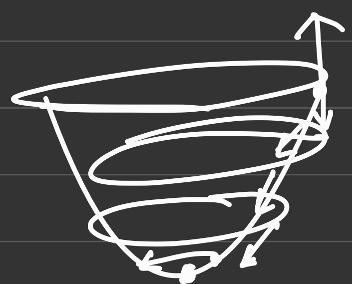
$D_{\vec{u}} f(x_0, y_0)$ when $\theta = 0°$

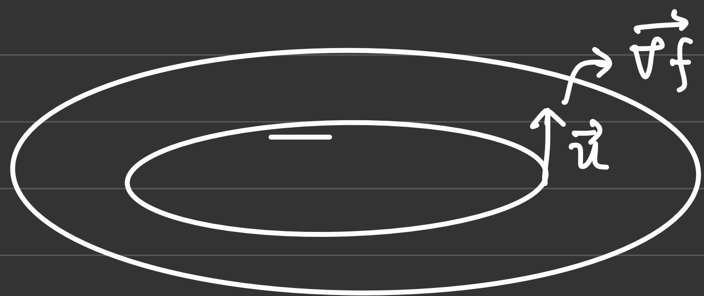
$-1 \le \cos\theta \le 1.$

$D_{\vec{u}} f(x_0, y_0) = \|\nabla f\|$ $\quad \theta = 0° \Rightarrow \vec{u}$ is along the direction of gradient $\vec{\nabla} f(x_0, y_0)$

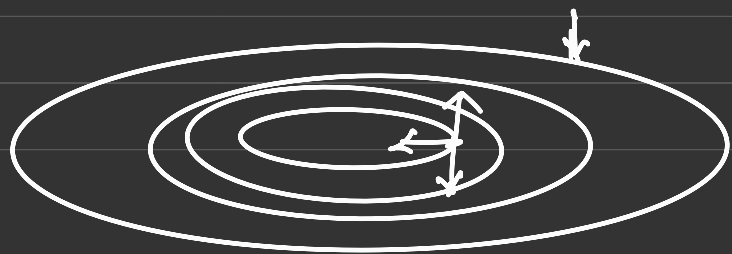


$\underline{\theta = 90°}$ No change





$\theta = 0° \rightarrow$ Direction of Steepest ascent



$\theta = 180° = -\|\nabla f\|$

Direction of steepest descent!

Max. Min Lagrangian

Least Squares & PCA as a max. problem.

$z = \underline{3x+5y}$ → Geometry → $3x+5y-z=0$ plane passing thro' the origin

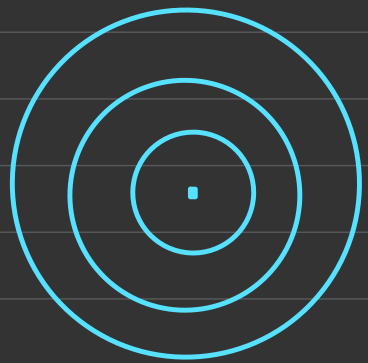
level curves corresp. to this

$$\left.\begin{aligned} z=1 &= 3x+5y \\ 0 &= 3x+5y \\ -1 &= 3x+5y \end{aligned}\right\} \text{Parallel lines.}$$

$x^2+y^2 = z.$

Surface :-

Level curves:



① Find a unit vector in the direction in which f increases most rapidly at P, find the rate of change of f at P in that direction

a) $f(x,y) = \dfrac{x}{x+y}$ $\quad P = (0,2)$

b) $f(x,y) = 4x^3y^2$ $\quad P = (-1,1)$

c) $f(x,y,z) = x^3z^2 + y^3z + z - 1$ $\quad P(1,1,-1$.

② Find $D_{\vec{u}} f(x_0, y_0) \rightarrow$ a) $f = 4x^3y^2$ $\quad (x_0, y_0) = (2,1)$

$\vec{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \rightarrow \hat{u}$

ⓑ $f(x,y) = e^x \cos y$, $P = (0, \pi/4)$, $\vec{u} = \begin{pmatrix} 5 \\ -2 \end{pmatrix} \hookrightarrow \hat{u}$.